So many SPECIES

Evolution and variation

The reason why there are so many species is evolution. Evolution is a process of change that happens gradually over millions of years. Tiny alterations in the way an organism looks or behaves may give it an edge over other members of its species in its ability to survive. Sometimes new features are adapted and passed down through succeeding generations until the descendent looks or behaves differently from its ancestor. Then it may be classed as a new species.

I could be an elephant one day!

Moeritherium

Gomphotherium

Natural selection

Adaptations make it easier for organisms to make use of a unique place in their ecosystem. That way, similar species can live together without competing for the same resources. Two related birds might not be able to live in the same tree if they both need the same type of food, but if one has a short bill suitable for eating insects and the other has a sharp bill for eating fruit then they can share a habitat. If they both fed on insects then the species that was better at catching them would eventually push the other species out. This process, where the best adapted survives, is called natural selection.

Sharp-billed fruit eater

Green honeycreeper

Short-billed insect eater

Scarlet tanager

They didn't find me until 2008, and haven't yet properly classified me!

This long-nosed frog is also called "Pinocchio". Scientists don't know if it is a new species of *Litoria* or not.

Voyage of discovery

Even though humans have visited everywhere on land there are still many places that haven't been properly explored. Even less is known about the ocean because of the difficulties of exploring in deep water. Scientists think that there could be as many as a million species in the sea and that we have only discovered 20 percent of them so far.